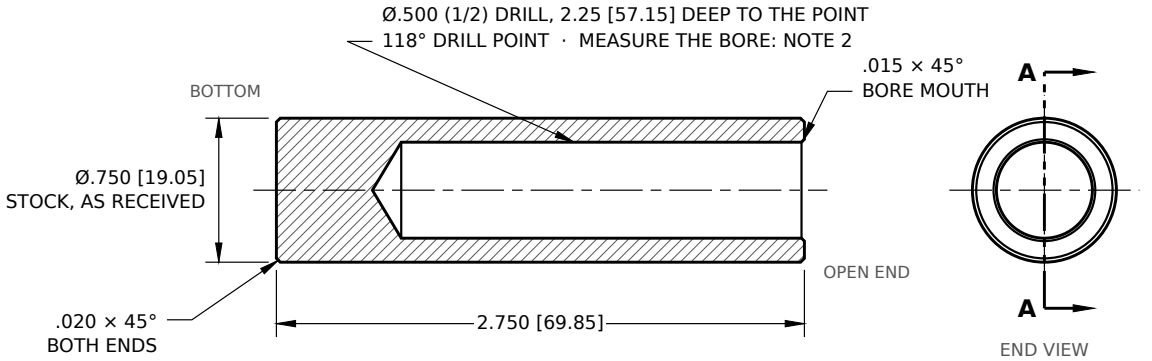
Charge cup and plug, as made

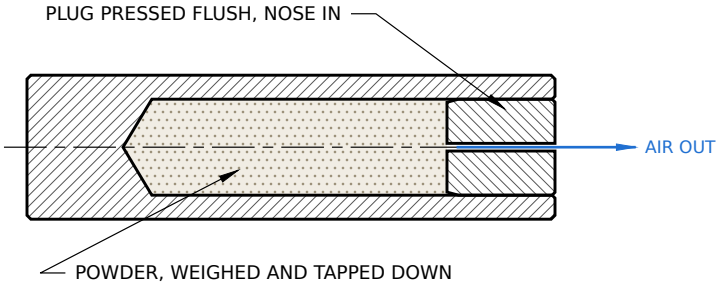
The sizes of the cups and plugs made for the first runs (#222), so new ones match them. Every part is drawn cut through the middle; hatching is metal.

REV	DESCRIPTION	DATE
A	FIRST ISSUE: THE AS-MADE SIZES (#222, PR #277)	2026-10-19



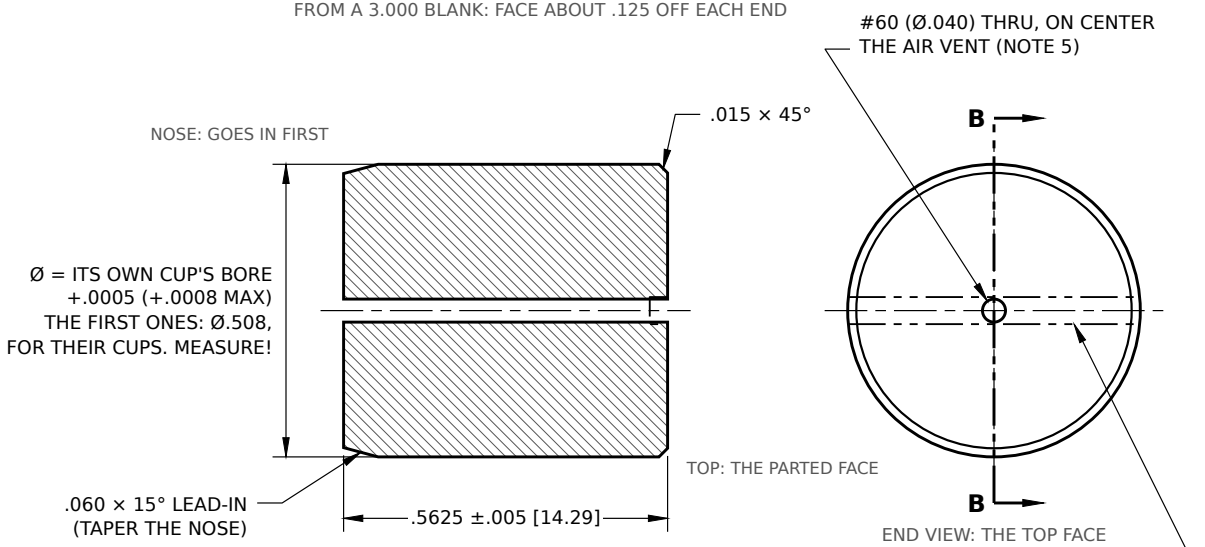
CUP · SECTION A-A · 1:1

FROM A 3.000 BLANK: FACE ABOUT .125 OFF EACH END



ASSEMBLED AND FILLED · REF · 1:1

STANDS PLUG-UP IN THE CRUCIBLE



PLUG · SECTION B-B · 3:1

ONE PER CUP, MADE AFTER ITS CUP

NOTES

1. MATERIAL: 6063 ALUMINUM ROD, Ø3/4 (McMASTER 1640T16), FROM THE 3.000 SAWN BLANKS. LEAVE THE OD AS RECEIVED.
2. MAKE THE CUP FIRST. MEASURE ITS BORE AT THE MOUTH WITH A TELESCOPING GAGE AND A MICROMETER (CALIPERS AREN'T FINE ENOUGH), AND TURN ITS PLUG TO THAT +.0005. NEVER OVER +.0008: A TIGHTER PLUG PERMANENTLY STRETCHES THE CUP WALL.
3. ONE PLUG PER CUP: THEY ARE NOT INTERCHANGEABLE. BAG EACH PAIR TOGETHER AND LABEL THE BAG. NO MARKER, SCRIBING OR STAMPS ON THE PARTS.
4. FACE THE OPEN END FIRST AND DRILL FROM IT, THEN FLIP THE BLANK AND FACE THE BOTTOM TO LENGTH. THE 2.25 DEPTH IS TO THE DRILL POINT: ZERO THE TAILSTOCK WHEN THE POINT TOUCHES THE FACE.
5. EVERY PLUG GETS THE #60 VENT. THE FURNACE IS PUMPED DOWN BEFORE MELTING, AND WITHOUT A VENT THE AIR UNDER THE PLUG HAS TO ESCAPE THROUGH THE POWDER.
6. CUT DRY OR WITH LIGHT OIL. NO WATER-SOLUBLE COOLANT, AND NO FILES, EMERY OR SCOTCH-BRITE: RESIDUE AND GRIT END UP IN THE MELT. BRUSH STEEL AND BRASS CHIPS OFF THE LATHE FIRST.
7. FLUSH THE BORE WITH 99% IPA, SWAB IT, FLUSH AGAIN AND LET IT DRY. WEIGH EACH PART TO 0.01 g AND WRITE THE MASSES ON THE BAG.
8. HYDRAULIC-PRESS RUNS ONLY: DRILL 31/64, THEN REAM OR BORE TO .500 SO THE BORE IS STRAIGHT AND SMOOTH, AND CUT THE CHANNEL. THAT PLUG SLIDES IN RATHER THAN PRESSING IN, AND ITS CLEARANCE ISN'T SET YET: ASK BEFORE MAKING ONE.

TWO VERSIONS: ANYTHING NOT LISTED IS THE SAME FOR BOTH

	STANDARD, AS MADE (SHOWN)	THIN WALL
FOR	ALL RUNS SO FAR	HIGH-POWDER RUNS (#264)
CUP BORE	Ø.500 (1/2) DRILL	Ø.625 (5/8) DRILL, AFTER THE 1/2
CUP WALL	.125	.0625
BORE DEPTH	2.25 TO THE POINT	2.25 TO THE POINT
PLUG LENGTH	.5625 (9/16)	.250 (1/4)
PLUG Ø	CUP BORE +.0005 (+.0008 MAX)	CUP BORE +.0005, NO MORE
MASS, CUP / PLUG (REF)	≈35 g / 5.0 g	≈25 g / 3.4 g

MASSES: 6063 AT 2.69 g/cm³, NOMINAL STOCK. THE BAR'S OD TOLERANCE MOVES THEM ABOUT 4 %.

BYU VERTICAL CLOUD LAB · rePOWDER ATOMIZER CHARGE

CHARGE CUP AND PLUG, Ø3/4 6063

DWG NO.	REV	SHEET	PROJECTION
AC-CUP-PLUG	A	1 OF 1	

SCALE: CUP 1:1, PLUG 3:1, ON LETTER AT 100 %

UNITS: INCH [mm]

MATERIAL · FINISH: 6063 ALUMINUM ROD, Ø3/4 (McMASTER 1640T16) · AS MACHINED, DEGREASED

UNLESS OTHERWISE SPECIFIED: .XX ±.01 .XXX ±.005 ANGLES ±1° BREAK SHARP EDGES .005-.015 DO NOT SCALE

DRAWN: CLAUDE (AI) FOR R. GUYMON, 2026-10-19

CHECKED:

SIZES AS MADE (#222); FIT AND THIN WALL FROM PR #232 · SOURCE: atomizer-charge/drawing/cup_plug.py